Encephalic Stroke Prediction Using MACS-Net

(Multi-modal Attention-driven Clinical and Spatial Network)

Abstract

Encephalic stroke, encompassing both ischemic and hemorrhagic cerebrovascular events, remains one of the leading causes of mortality and long-term neurological disability worldwide. Early and accurate prediction of stroke risk is critical for timely clinical intervention and prevention of irreversible brain damage. This study proposes MACS-Net (Multi-modal Attention-driven Clinical and Spatial Network), a novel deep learning architecture designed to integrate multi-modal patient data for robust encephalic stroke prediction. MACS-Net employs a dual-branch encoder: a Spatial Attention Module (SAM) that processes neuroimaging features to capture lesion-sensitive brain region activations, and a Clinical Feature Encoder (CFE) that encodes structured patient records encompassing key risk indicators including age, hypertension, cardiac conditions, blood glucose levels, BMI, smoking history, and prior neurological episodes. A cross-modal attention fusion layer dynamically aligns and integrates spatial and clinical representations, enabling context-aware risk stratification. To address class imbalance inherent in medical datasets, MACS-Net incorporates a focal loss objective combined with Synthetic Minority Oversampling Technique (SMOTE) augmentation, and hyperparameter tuning via cross-validation further enhances generalization. The proposed system achieves high accuracy, sensitivity, and AUC-ROC scores, demonstrating reliable differentiation between stroke and non-stroke cases. Key benefits include interpretable multi-modal attention maps for clinical explainability, low computational cost for clinical deployment, real-time risk scoring capability, and adaptability to Electronic Health Record (EHR) integration. This framework offers clinicians a data-driven decision support tool for proactive stroke risk stratification and early preventive care.